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Paleomagnetism of the Mississippi Valley-Type Newfoundland Zinc Deposit: Evidence for Devonian Mineralization and Host Rock Remagnetization in the Northern Appalachians

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Newfoundland Zinc Mines (NZM) are exploiting sphalerite-rich Mississippi Valley-type (MVT) zones in platformal carbonate rocks of the Humber Zone in western Newfoundland. The zones are stratabound within the upper part of the Lower Ordovician St. George group. Geologic studies in the mine area simply require ore genesis to postdate the Early to Middle Ordovician Taconic orogeny. Thermal and alternating field (AF) demagnetization reveal that both the ore and adjacent host rocks at the NZM have similar characteristic remanence magnetizations (ChRM) with normal polarity. The ChRM direction from both ore and host rocks is at 9°/-30° (declination/inclination) ($N = 16$ sites, $\alpha_{95} = 6°$, $k = 46$) which gives a pole position of 22°N, 113°E ($dp = 3°$, $dm = 6°$). This pole falls on the boundary between the Middle and Late Devonian on the North American apparent polar wander path and corresponds in time to the Acadian orogeny. Paleomagnetic breccia tests show that the host rocks were remagnetized after the Middle Ordovician and that this remagnetization coincides with or postdates MVT mineralization. The ChRMs paleoinclination indicates that both remagnetization and mineralization are not younger than Late Devonian. The similar magnetization characteristics of the adjacent and remote host rocks in the Humber Zone and the authigenic nature of the magnetic carriers revealed by scanning electron microscopy with energy dispersive analysis suggest that the NZM host rock has a secondary magnetization. From the pervasive nature of both the secondary remagnetization and the trace MVT mineralization in the northern Humber Zone, we argue that both events are coeval. We suggest that Acadian orogenic uplift forced hot basinal brines into the porous and fractured Ordovician shelf carbonates of the northern Humber Zone.

INTRODUCTION

Recent paleomagnetic studies of Mississippi Valley-type (MVT) deposits have provided independent constraints on their ages of mineralization and thus for their genetic models [*Beales et al.*, 1974; *Wu and Beales*, 1981; *Wisniowiecki et al.*, 1983; *Bachtadse et al.*, 1987; *Pan et al.*, 1990; *Symons and Sangster*, 1991, 1992]. For example, several studies of MVT deposits in the midcontinent of North America have found that the ores and adjacent host rocks retain a late Paleozoic Kiaman magnetization which is coincident in time with the Alleghenian or Ouachita orogenies in those areas. From their magnetization, mineralization, and geologic characteristics, these MVT deposits are interpreted to have formed from hot, migrating, intraformational fluids that were driven by orogenic uplift. Thus the paleomagnetic results support gravity-driven fluid-flow genetic models for MVT deposits in the midcontinent [*Garven and Freeze*, 1984*a*, *b*; *Oliver*, 1986; *Leach and Rowan*, 1986]. Furthermore, these paleomagnetic studies of MVT districts provide an insight into the relationship between wide-spread Kiaman remagnetization processes and tectonically driven hot fluids in platformal carbonate rocks [*McCabe and Elmore*, 1989].

We sampled ore and adjacent host rocks from the MVT deposits of the Newfoundland Zinc Mines (NZM) in western Newfoundland to attempt to date the mineralization paleomagnetically (Figure 1). The NZM deposits are of interest for several reasons. First, they are situated in the northern Appalachian orogenic belt in contrast to the MVT deposits in the undisturbed carbonate platform of the midcontinent of North America. The latter are genetically related to the Alleghenian or Ouachita orogenies whereas the rocks in the Humber Zone are virtually unaffected by these orogenies [*Williams*, 1980; *Cawood and Williams*, 1988]. Second, the Early Ordovician St. George group, the host rock for the NZM deposits, is relatively well exposed along much of western Newfoundland and has been the subject of recent paleomagnetic studies [*Deutsch and Rao*, 1977; *Prasad*, 1986; *Deutsch and Prasad*, 1987; *Hall and Evans*, 1988; *Beaubouef et al.*, 1990]. This allows comparison of the NZM ore results with those from host rocks, both nearby and at some distance from the ores, to examine the relationship between magnetization and mineralization, and thereby to better determine the age of mineralization for the deposit. Third, the NZM was studied paleomagnetically by *Beales et al.* [1974]. They defined an Early Ordovician or syngenetic age of mineralization, suggesting the NZM deposit was formed during the Taconic orogeny. This conclusion is not supported by the geologic evidence for a post-Taconic origin or by current epigenetic models for MVT deposits. Therefore the need to restudy the deposit and host rocks using more modern paleomagnetic methods was evident.

GEOLOGY

The NZM deposit is a sphalerite-rich MVT deposit. The ore lenses are strata-bound within the upper part of the Lower Ordovician St. George group, part of an autochthonous or parallochthonous sequence of Cambro-Ordovician shallow-water platformal sedimentary rocks which lies unconformably on the late Precambrian basement of the Grenville province (Figure 2). Nearby an allochthonous thrust sheet assemblage of Middle Ordovician deep-water sedimentary rocks and ophiolites covers these autochthonous shallow-water rocks. Together these lower Paleozoic autochthonous and allochthonous sequences

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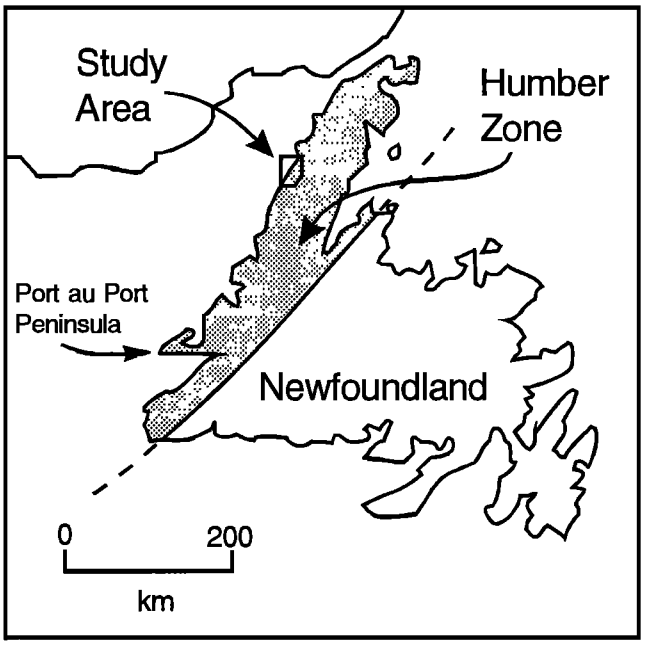


Fig. 1. Map of Newfoundland showing the Humber Zone and study area.

comprise a distinct tectono-stratigraphic terrane known as the Humber Zone, one of five such subterranes in the northern Appalachians of Newfoundland [*Williams*, 1979, 1984].

The St. George group is a succession of marine subtidal and peritidal carbonates that originally accumulated on the outer part of a gradually subsiding, low-latitude continental margin [*Knight and James*, 1987; *Williams and Stevens*, 1974; *Williams et al.*, 1987]. The St. George group conformably overlies rocks of the Upper Cambrian Port au Port group. The former is composed, in ascending order, of the Watts Bight, Boat Harbour, Cotache and Aguathuna formations (Figure 3) which reflect sea level changes on the Early Ordovician continental shelf. Of these, only the Cotache and Aguathuna formations were sampled. The Cotache formation is a sequence of shallow, subtidal carbonate rocks composed dominantly of limestone that are variably affected by dolomitization. The upper part of the Catoche is extensively altered at Daniel's Harbour, where medium to coarse, grey/white, epigenetic dolomites host extensive zinc mineralization [*Collins and Smith*, 1975; *Lane*, 1984, 1990]. The overlying Aguathuna formation comprises a series of finely crystalline, grey peritidal dolostones with minor interbedded limestones. The Aguathuna formation was formed during the initial stages of continental margin flexure, influenced by obduction at the onset of the Taconic orogeny. The St. George group is overlain unconformably at some, but not all,

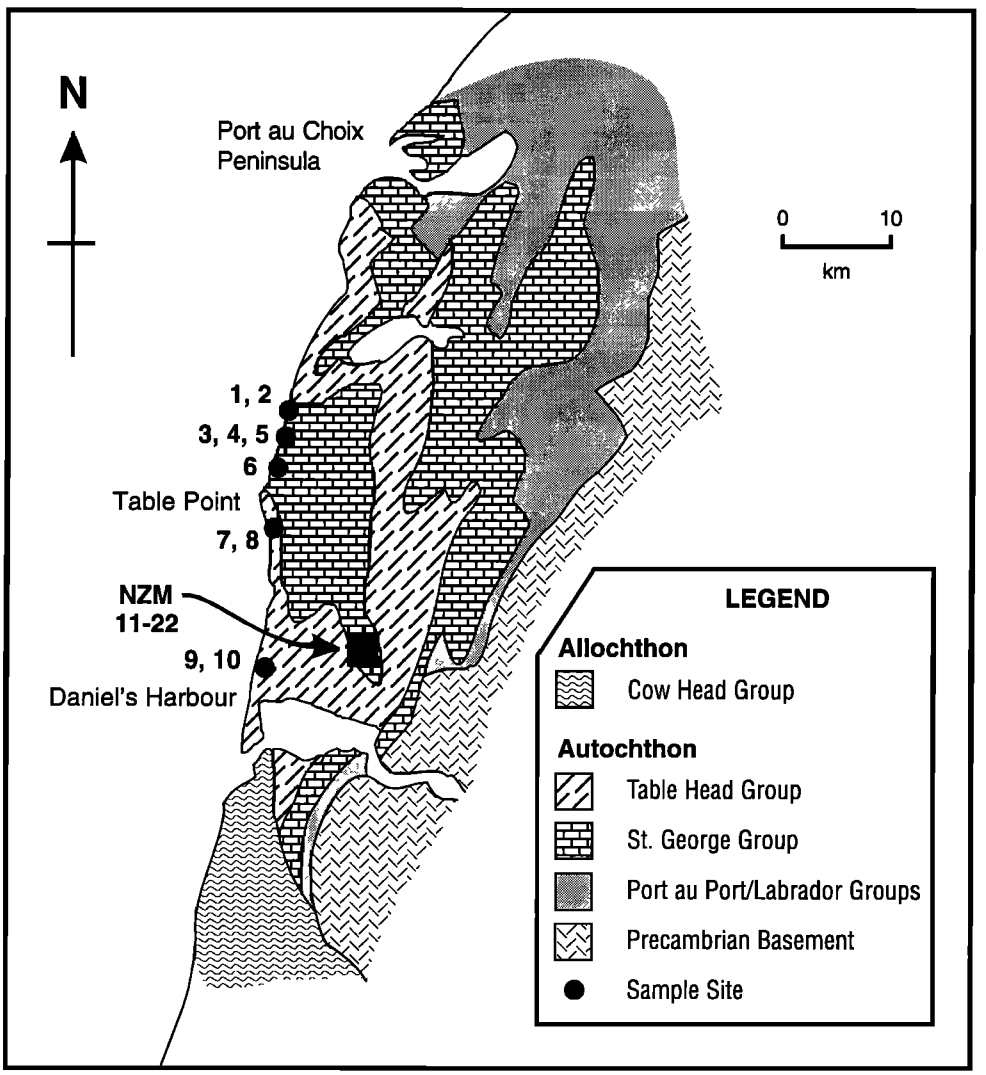


Fig. 2. Geologic map of the Daniel's Harbour area showing locations of sampling sites and Newfoundland Zinc Mines (NZM).

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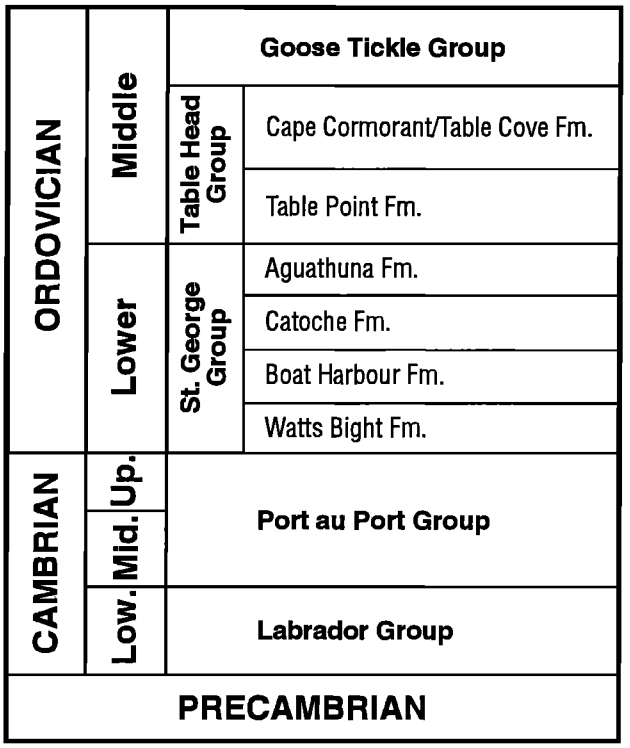


Fig. 3. Generalized stratigraphic sequence of the study area.

localities by well-bedded limestones of the Middle Ordovician Table Head group. The Table Head group was deposited in a progressively deepening environment that records the collapse of the continental platform during the Taconic orogeny [*Klappa et al.*, 1980; *Stenzel et al.*, 1990]. It is subdivided into three formations: the Table Point, Table Cave, and Cape Cormorant [*Stenzel et al.*, 1990] (Figure 3).

In the mine area, the upper St. George group rocks and associated zinc deposits are almost flat-lying. The mine encompasses a 16 km^2 area within which more than 15 ore lenses are scattered [*Lane*, 1990] (Figure 4). These ore lenses were mined in open pits and underground. The sphalerite was precipitated with epigenetic, coarse-crystalline, white sparry dolomite (saddle dolomite) which envelopes the sphalerite lenses. The sphalerite may cement pores, fill veins, or replace previously formed dolomite. Massive mineralized zones contain 60% to 80% sphalerite but may exceed 90% at some localities. *Lane* [1990] reported fluid inclusion data for both sphalerite and adjacent dolomite samples. He found final melting temperatures for both that range from -22°C to -25°C, corresponding to salinities of 22 to 25 equivalent weight percent NaCl. The homogenization temperatures of sphalerite inclusions vary widely, ranging from 90° to 180°C with a mean of 140°C [*Lane*, 1990].

SAMPLING

Four oriented block samples were collected from each site except for the sites used for breccia and conglomerate tests, where five to six samples were collected. Of the 22 sites sampled in this study (Figure 2), 11 were ore sites from both open pits and underground workings that include disseminated and massive sulphides (Figure 4). These ore sites represent deposits at three stratigraphic levels. Four other sites were sampled for remanence stability tests to constrain the age of magnetization. Of these four, two sites with a total of 10 blocks were sampled from late-stage breccias. Each block represents an unmineralized dolostone breccia fragment of 0.2 to 0.4 m diameter. The fragments are cemented by saddle dolomite and sulphides. The bedding in the sampled fragments showed that they had been significantly rotated. The other two sites, with a total of 11 blocks, were sampled from an unmineralized syndepositional dolostone breccia in the lower Catoche formation from well below the ore horizons. The host rocks were sampled near the ore bodies. They include three sites in completely dolomitized Catoche formation at the same stratigraphic levels as the ore bodies. Another site was collected in the overlying

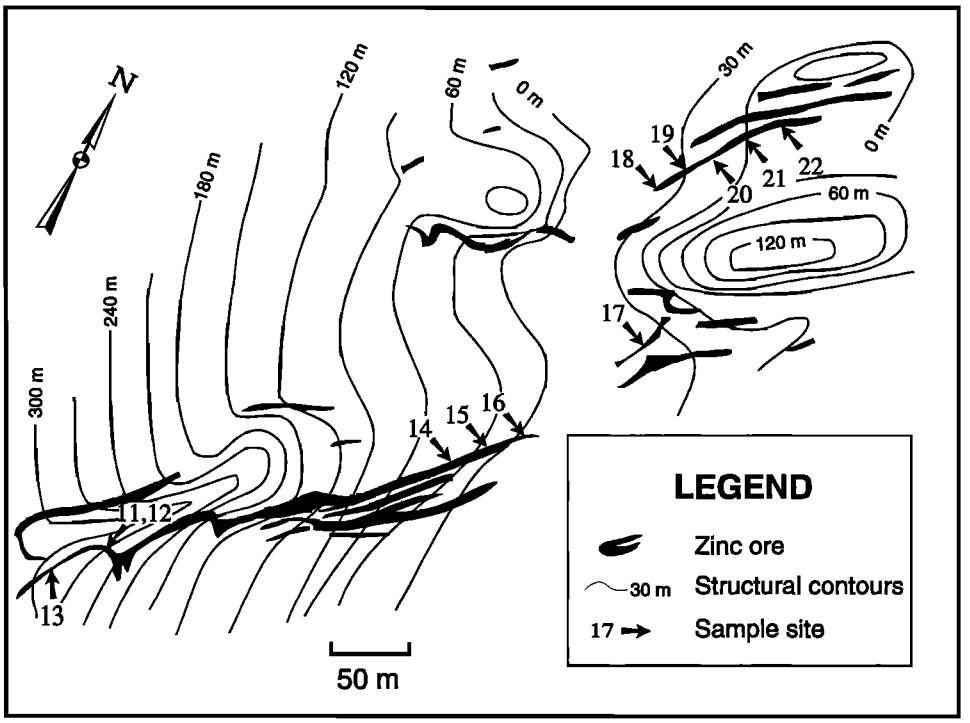


Fig. 4. Map of the ore zones at NZM with sampling sites. Structural contours indicate the top of Cotache formation relative to a datum plane at 106 m above sea level after Lane [1990].

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Aguathuna formation. In addition, two sites were sampled in the folded Table Point formation of the Table Head group about 20 km away from the mine to provide a paleomagnetic fold test. Finally, 18 limestone clasts with diameter of approximately 0.25 m were sampled from two conglomerate sites at Daniel's Harbour to provide a paleomagnetic conglomerate test. The conglomerate beds were mapped as part of the Cape Cormorant formation of the Table Head group at the time of collecting, but are now recognized as a separate unit in the Goose Tickle group [*Stenzel et al.*, 1990] (Figure 3).

EXPERIMENTAL TECHNIQUES

Paleomagnetism

Generally, about four specimens with a 2.5-cm diameter and 2.2-cm length were drilled from each block sample. Prepared specimens were stored and measured in a magnetically shielded room with an ambient magnetic field of about 100 nT. All remanence measurements were made using an automated two-axis CTF Systems Inc. DRM-420 cryogenic magnetometer with a noise level of less than 4×10^{-6} A/m. One pilot specimen from each block was demagnetized in 16 steps using alternating fields (AF) up to 160 mT. A second pilot specimen was thermally demagnetized in 10 steps up to 600°C or until the magnetization behavior became erratic. The comparison of thermal and AF demagnetization for specimens from the same block or site showed that their magnetization behaviors are similar. Therefore, most remaining specimens were AF demagnetized in at least six steps over a coercivity spectrum of >30 mT. A few specimens with residual high magnetic coercivities were thermally demagnetized. The demagnetization data were evaluated using principal component analysis [*Kirschvink*, 1982] with the maximum angular deviation (MAD) required to be less than 10° for acceptance. Mean directions were calculated using standard *Fisher* [1953] statistics. Further description of the experimental methods used in this study are given by *Sangster and Symons* [1991] and *Symons and Sangster* [1993].

Rock Magnetism

The mineralogy of the magnetic carriers was examined using the acquisition of saturation isothermal remanence magnetization (SIRM) and its subsequent AF demagnetization. Eleven specimens were selected from the various rock units with a variety of demagnetization behaviors. Each specimen was magnetized using a Sapphire Instruments SI-6 pulse magnetizer in nine steps up to 900 mT. The SIRM was then AF-demagnetized in five steps up to 50 mT. In addition, magnetic extracts were separated from ten samples to observe grain morphology and chemistry using scanning electron microscopy (SEM) with energy dispersive analysis (EDA). The samples were prepared by crushing and grinding whole-rock specimens using nonmagnetic equipment. The extracts were obtained by slowly pumping a dilute water suspension of each ground specimen through a magnetic separator [*Pan et al.*, 1990].

RESULTS

Paleomagnetic Results

The natural remanent magnetization (NRM) directions of all specimens are scattered, but most are inclined downward to the north at an intermediate angle. The NRM intensities range from 2.3×10^{-3} to 3.5×10^{-5} A/m. No significant difference was found between the NRM intensities of the host rock and the ore. Most specimens exhibit similar behavior upon AF and thermal step demagnetization, indicating that both methods are effective at removing the viscous remanent magnetization (VRM) to isolate the characteristic remanent magnetization (ChRM). Most specimens are characterized by relatively simple behavior upon demagnetization, revealing a two-component NRM regardless of lithologic differences (Figures 5 and 6).

Upon thermal demagnetization, the first component is usually removed at temperatures below 350° to 450° C. This low unblocking temperature component is most commonly directed steeply downward to the north and comprises 60-80% of the initial NRM intensity. It is interpreted to be a modern VRM acquired in the present Earth's magnetic field (EMF). In most cases a rapid change in direction follows and the remaining NRM decays linearly toward the origin. This high unblocking temperature component is removed at temperatures up to 525° - 550°C (Figures 5*a*, 5*c*, 5*e*, 6*a*, and 6*c*). The range of unblocking temperatures suggests that the ChRM carrier is magnetite. During thermal demagnetization many specimens, particularly those of massive sulphide, show increasing NRM intensities with inconsistent NRM directions at unblocking temperatures above about 400° to 500°C. These features indicate the conversion or production of new magnetic minerals because of oxidation. Such erratic data points were rejected, and the preceding lower-temperature linear decay segment was extrapolated to the origin to determine the ChRM direction. For most specimens the ChRM is directed upward to the north at a moderate inclination and it was found at all sites except site 6 in dolostone, sites 9 and 10 in conglomerate clasts, and site 22 in mineralization (Table 1). Upon AF demagnetization, the VRM component is removed initially by field intensities of 20 - 25 mT and then the ChRM component is removed progressively up to 130 - 160 mT (Figures 5*b*, 5*d*, 5*f*, 6*b* and 6*d*). Because this high-coercivity ChRM is unblocked up to about 550°C, the remanence carrier in the specimens is thought to be fine-grained magnetite.

Vector plots for samples from sites 6 and 22 indicate either (1) the appearance of a steep univectorial remanence component, which is close to the present EMF direction, that is likely caused by modern weathering (Figure 7*a*) or (2) multiple components with erratic directions. Therefore the data for these two sites were rejected from further analysis.

Stability Tests

In this study, clasts were sampled from a syndepositional intraformational dolostone breccia at sites 1 and 2 [*Lane*, 1984, 1990] and from a solution-collapse breccia in which the dolostone breccia fragments are cemented by ore-stage saddle dolomite and sulphides at sites 11 and 12 [*Lane*, 1990]. The magnetization behavior and ChRM direction of all breccia fragments are indistinguishable from those of the limestone or dolomite host rocks or the mineralized samples (Figures 6 and 8). This similarity of ChRM directions in breccia, unbrecciated host rocks, and ores clearly indicates that the ChRM is secondary and postbrecciation.

Conglomerate clasts were collected from sites 9 and 10 in the Daniel's Harbour member of the Goose Tickle group. The clasts' ChRM directions are not similar to those of the ore, host rock, and breccia units. Some clasts retain a steep univectorial component (Figure 7*b*). However, most clast specimens did not reach a stable endpoint before they become too weak to measure (Figure 7*c*) but their great circle trajectories do diverge from the present EMF direction to indicate the removal of a low-coercivity modern VRM (Figure 9*a*). Combining the normals to the remagnetization planes with the directions given by the stable endpoint of some clasts [*Bailey and Halls*, 1984], a mean direction of 344°/71° (declination/inclination) ($\alpha_{95} = 13°$) is obtained that is statistically the same as the present EMF direction of 333°/73° (Table 1). This implies that the clasts are contaminated with an unremoved modern VRM overprint. As these clasts were collected from a tidal flat outcrop, it is likely that they have been weathered during Recent time.

The ChRM isolated at sites 7 and 8 in the folded Table Point formation is also very similar to that of the host rock, ore, and breccia units (Figure 5). Unfortunately, the ChRM direction is almost parallel

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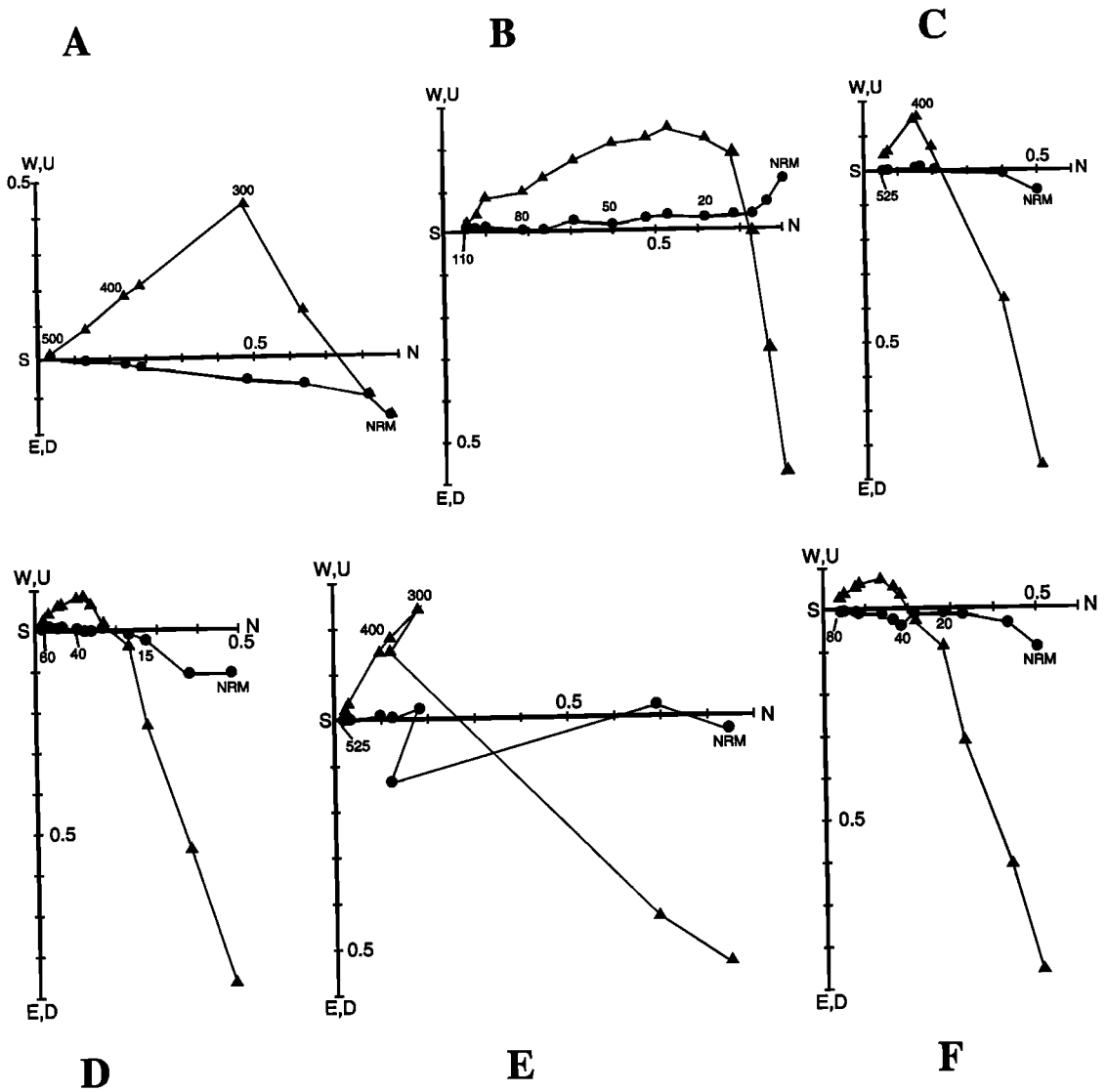


Fig. 5. Orthogonal demagnetization diagrams of representative specimens showing two-component magnetizations isolated by AF and thermal demagnetization in (*a*) and (*b*), ore (NRM intensity: 5.1×10^{-4} A/m and 1.4×10^{-3} A/m); (*c*) and (*d*), host dolostone (2.3×10^{-3} A/m and 2.2×10^{-3} A/m); (*e*) and (*f*), limestone (1.2×10^{-3} A/m and 1.4×10^{-3} A/m). Circles and triangles denote projections on the horizontal and vertical planes, respectively. The axis unit is normalized to the initial NRM intensity. Treatments are in mT and °C for AF and thermal demagnetization, respectively.

to the fold axis, so that there is only slight increase in the precision parameter (*k*, *Fisher* [1953]) upon unfolding and therefore the fold test is inconclusive.

Rock Magnetism

The SIRM acquisition curves for all test specimens from the various lithologies show that the IRM increases rapidly and mostly reaches saturation by 300 mT. The IRM curves fall between the type curves for multidomain (MD) and single-domain (SD) magnetite, indicating that the magnetic carrier is fine-grained magnetite (Figure 10). AF demagnetization of the SIRM shows more precisely that the magnetite is SD or pseudo-single domain (PSD) in character (Figure 10). These results are consistent with demagnetization behavior. It is noteworthy that most of the normalized intensity decay curves from AF and thermal step demagnetization are nearly identical, indicating that the coercivity or unblocking temperature spectra for most specimens from the various lithologies are quite uniform. Thus the SIRM acquisition and AF demagnetization tests indicate that the magnetic carriers in all rock types have a very similar grain-size distribution pattern [*Stacey and Banerjee*, 1974; *Heider et al.*, 1987], implying a common origin for remanence carrier in all rock types.

Scanning Electron Microscopy

SEM observations of the magnetic mineral extracts show variations in grain morphology between different lithologies, but irregularly shaped blocky grains are the most abundant magnetic particles in all extracts (Figures 11*a* and 11*b*). The size of these blocky grains ranges from 1 - 100 m. In some cases Fe is the only cation detected by EDA. In other cases Fe is found with a significant concentration of Ca, Mg, Si, S, Al, and Zn. The presence of Ca, Al, Si, and Mg may be attributed to clays trapped in the ferric oxide coating on some magnetic particles [*Suttill et al.*, 1982]. It is also probable that very small magnetic particles are trapped in sulphide, dolomite or calcite crystals [*Symons and Sangster*, 1991]. No Ti was found in the extracts, indicating relatively pure magnetite.

In the extracts, some textural features were found which characterize an authigenic origin. The extract from the syndepositional breccia contains magnetic aggregates of rounded, separated, fine-grained

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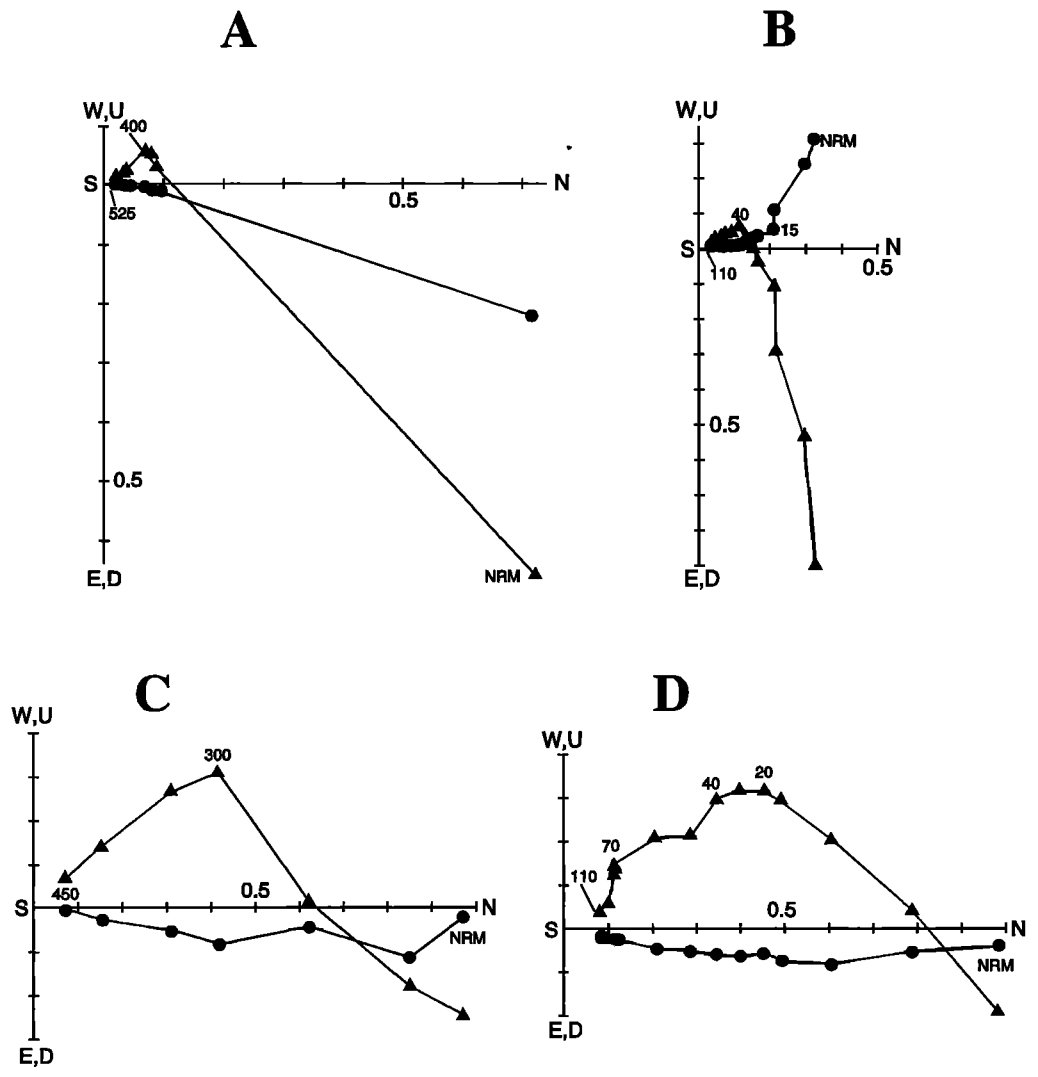


Fig. 6. Orthogonal demagnetization diagrams of representative specimens showing two-component magnetizations isolated by AF and thermal demagnetization in (*a*) and (*b*), syndepositional breccia clasts (1.1×10^{-3} A/m and 2.3×10^{-3} A/m); and (*c*) and (*d*), late stage breccia clasts (1.1×10^{-3} A/m and 1.2×10^{-3} A/m). Symbols are the same as in Figure 5.

particles, each one of which is approximately 0.1 - 1 μm in diameter (Figure 11*c*). In such extracts, Fe is a dominant cation but Ca and Si are also present. Similar aggregates of very fine-grained particles were also found in the ore extracts (Figure 11*d*). In this case, the individual particles are perfect spheroids with Fe being the only cation detected by EDA. These individual spheroids are sometimes clustered into aggregates or interwoven into worm-shaped aggregates. A couple of extracts from the ore samples also contain aggregates of needle-shaped particles (Figure 11*e*) with Fe being the only detected cation in them. Occasionally, SEM revealed a few small individual spheroids of 1 - 2 μm in diameter from both ore and host rock (Figure 11*f*). These spheroids contain Fe as the only element and resemble those found in the midcontinent MVT deposits of North America [*Wisniowiecki et al.*, 1983; *Bachtadse et al.*, 1987; *Pan et al.*, 1990].

We can infer that the magnetic particles revealed by SEM are the significant remanence carriers for the following reasons. First, almost all of the magnetic particles are seen to be in the SD or PSD size range, which is consistent with the SIRM and AF demagnetization results. Second, the extracts were prepared from whole-rock samples so that their magnetic particles are representative of the range of actual remanence carriers in the rocks. Thus we conclude from the textural features revealed by SEM that most magnetic particles, if not all, are of authigenic origin. Also the absence of Ti in the extracts implies the absence of detrital magnetite grains because igneous or metamorphic origins commonly yield detrital titanomagnetite.

DISCUSSION

Unit Mean Direction

The 16 accepted site mean ChRM directions are statistically indistinguishable when grouped by lithology (Figure 12; Table 1). Therefore, all 16 are combined to obtain a unit mean of 9°/-30° (N = 16, $\alpha_{95} = 6°$, $k = 46$). This unit mean direction is significantly different from that of 357°/-25° (α_{95} = 7°) reported by *Beales et al.* [1974], although they also found normal-polarity ChRM directions for the ore and host rocks (Figure 13). This difference in mean directions appears to be caused by the incomplete AF demagnetization of their samples by *Beales et al.* [1974]. We found that isolation of the ChRM for most specimens requires AF demagnetization above 25-30 mT (Figures 5 and 6). Even though *Beales et al.* [1974] used five AF field intensities up to 30 mT, they used directions from the 10 mT step for their pole calculation. We found that the ChRM intensities for most specimens are only about 20-40% of the initial NRM intensities of 0.04-2.0×10

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TABLE 1. Site Mean Results

Site	Lith., Strat.	Specimens N_o/N	Dec./Inc., deg	k	α_{95}, deg
1 and 2	B1, Cl	9*	009/-22	52	7
3	L, Cu	12/7	005/-28	53	8
4	L, Cu	12/8	012/-38	130	5
5	D, Cu	12/12	011/-44	50	6
6†	D, A	12	-	-	-
7	L, TP	12/11	006/-50	66	6
8	L, TP	12/12	010/-33	56	6
9 and 10	C, DHM	18,4‡	344/ 71	15	13
11 and 12	B2, Cu	9*	020/-36	12	16
13	Omd, Cu	12/3	023/-36	13	16
14	Ohi, Cu	12/4	351/-12	27	18
15	Omd, Cu	12/9	016/-34	36	9
16	Omd, Cu	12/4	002/-34	13	8
17	Olo, Cu	12/11	009/-32	143	4
18	Omd, Cu	12/10	006/-31	50	4
19	Omd, Cu	12/12	011/-26	40	7
20	Olo, Cu	12/11	003/-15	31	8
21	Ohi, Cu	12/10	006/-19	28	9
22†	Omd, Cu	12	-	-	-

Notes: Lith, Lithology; Strat, stratigraphy; L, limestone; D, dolomite/dolostone; C, conglomerate; B1, syndepositional breccia; B2, epigenetic ore-stage breccia; Olo, Omd, Ohi, low (<10% ZnS), medium (10-60%), and high (>60%) ore grade; Cl, Cu, lower and upper Catoche formation; A, Aguathuna formation; TP, Table Point formation; and DHM, Daniel's Harbour member. N_o/N is the number of specimens measured and used for the mean. Mean direction: Dec., declination; Inc., inclination; α_{95}, radius of 95% confidence; k, precison parameter; dp, dm, radii of oval of 95% confidence [*Fisher*, 1953]. Unit mean: Dec = 9°, Inc = -30°, $k = 46$, $\alpha_{95} = 6°$, $N = 16$; pole position: 23°N, 113°E, $dp = 3°$, $dm = 6°$.

† Sites 6 and 22 gave incoherent ChRM directions.

* The number of clasts.

‡ Conglomerate mean combines four clast mean ChRM directions with eight mean normals to remagnetization planes [*Bailey and Halls*, 1984].

[3] A/m after the VRM was removed. This weak 10^{-5} A/m measurement range for the ChRM was beyond the sensitivity of the spinner magnetometer used by *Beales et al.* [1974], thus causing increased scatter in their directions for AF fields above 10 mT. Evidence for their ChRM direction being contaminated with VRM is suggested by the fact that their direction has a shallower inclination than our direction because an upward ChRM direction would be made more shallow by the addition of a modern downward VRM component.

Maximum Age of Remagnetization and Mineralization

As noted earlier, rocks of the St. George group have been examined in several paleomagnetic studies in the Port au Port area about 260 km southwest of NZM [*Deutsch and Rao*, 1977; *Deutsch and Prasad*, 1987; *Hall and Evans*, 1988; *Beaubouef et al.*, 1990]. Although each study found a slightly different direction for the St. George group, all report a moderately inclined ChRM direction of reversed polarity which the authors argued to be a primary or early diagenetic remanence on the basis of the determined pole position (Figure 13). *Hodych* [1989], using a combination of paleomagnetic breccia and fold tests on rocks from the overlying Ordovician Table Point limestones, has shown that this remanence was acquired during or after diagenesis but before Devonian folding. Conversely, the ChRM direction for the NZM has a normal polarity that is not statistically antiparallel to any of the directions from the Port au Port area (Figure 13), supporting our interpretation that the NZM magnetization is secondary in origin rather than a primary Early Ordovician magnetization.

A post-Early Ordovician age for the ChRM is also confirmed by the two negative breccia tests and the authigenic appearance of magnetic carriers revealed by SEM with EDA. The syndepositional intraformational breccia test shows that the magnetization of the breccia clasts was acquired after incorporation into the breccia, indicating that their magnetization is secondary. Stratigraphic relationships in the NZM area show that the breccia was formed below the depositional surface during karstification in the Early to early Middle Ordovician [*Lane*, 1990]. This breccia test constrains the remagnetization ChRM to be at least post-early Middle Ordovician. The late-stage breccia test more specifically requires that remagnetization be coeval with or postmineralization because the individual breccia clasts are cemented by ore-stage saddle dolomite and sulphides. The fact that both types of breccia clasts carry the same ChRM direction as the host rocks indicates that all of these lithologies were remagnetized, probably when the sulphide ores of the NZM were formed after the early Middle Ordovician.

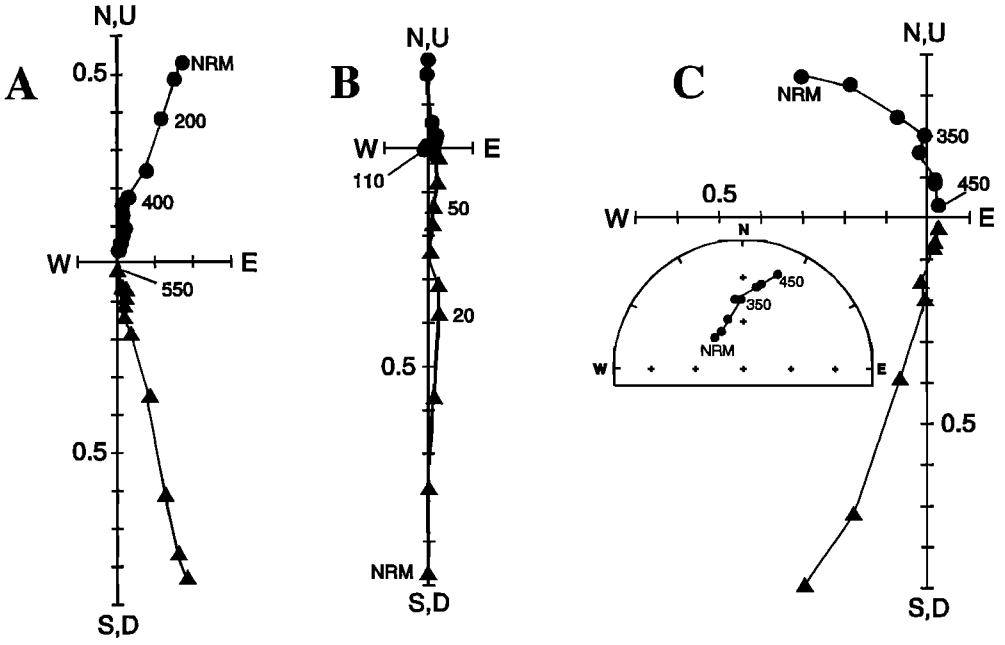


Fig. 7. Orthogonal demagnetization diagrams of typical specimens showing (*a*) steep single magnetization revealed by AF demagnetization from site 22 (2.2×10^{-3} A/m); (*b*) single-component magnetization isolated by AF demagnetization for a conglomerate clast (4.4×10^{-4} A/m); (*c*) NRM trajectory following a remagnetization plane to an endpoint by thermal demagnetization (2.6×10^{-4} A/m). Other symbols are the same as in Figure 5.

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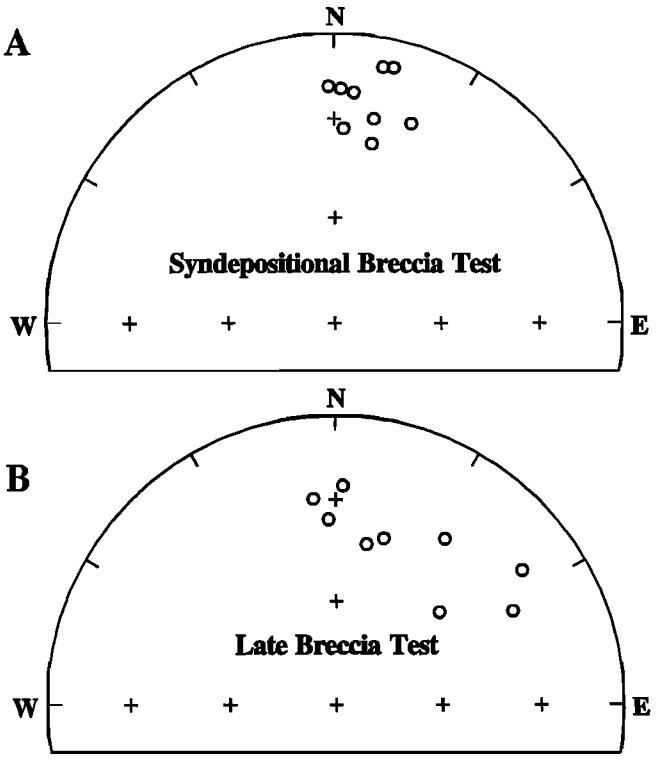


Fig. 8. Equal-area stereographic projection of block mean directions after demagnetization for (*a*) syndepositional breccia clasts; (*b*) late-stage breccia clasts. Open symbols represent the projections on the upper hemisphere, and solid symbols on the lower hemisphere.

In the Daniel's Harbour area, trace to minor MVT mineralization is widely distributed in the autochthonous carbonates and in the allochthonous Cow Head group [*Coniglio*, 1985]. Tectonic studies show that the allochthonous rocks were emplaced during the Taconic orogeny, a 25-million-year time interval during the Early to Middle Ordovician [*Stevens*, 1970; *Cawood and Suhr*, 1992]. Therefore the tectonic relationships also require that the MVT mineralization postdate the early Middle Ordovician and Taconic orogeny.

Minimum Age of Remagnetization and Mineralization

The unit mean ChRM direction for the flat-lying NZM rocks corresponds to a pole position of 22°N, 113°E ($dp = 3°$, $dm = 6°$). This pole falls on the boundary between the Middle and Late Devonian segments of the North American apparent polar wander (APW) path of *Van der Voo* [1990] (Figure 14). If the remagnetization of the host rocks was substantially younger than the Late Devonian, then it should be recorded mostly as a change in paleolatitude to give a different remanence inclination. Tectonic rotation about a vertical axis of the NZM area as part of a thrust sheet could not cause a younger remagnetization to appear to give this age. Therefore the paleomagnetic pole provides a minimum age for the MVT mineralization and remagnetization of the host rocks.

Massive, "almost pure" sphalerite samples from the NZM deposits formed as botryoidal growths filling cavities, and they exhibit the same magnetization characteristics and ChRM direction as those from the low-grade sphalerite ore, dolostone, and limestone. Authigenic magnetite is seen by SEM with EDA in these ore samples, as it is for the remagnetized host rocks, implying that the magnetization is a chemical remanence associated with mineralization. These features also set a minimum age of Late Devonian for remagnetization and mineralization at NZM.

Acadian Orogeny and Ore Genesis

In the NZM area, the conodont color alteration index (CAI) value is approximately 2.5 which corresponds to a paleotemperature of less than 200°C [*Nowlan and Barnes*, 1987]. Fluid inclusions in sphalerite indicate that the MVT deposits were formed in the 90° to 180°C temperature range from homogenization temperatures and that subsequent reheating has not surpassed this range [*Bodnar and Bethke*, 1984]. Thus the maximum postmineralization temperature from either burial or a hot brine is under 200°C. With maximum unblocking temperatures for magnetite in the ores and host rocks of >550°C, a burial temperature of >400°C for more than 1 Ga would be required to reset their magnetization [*Pullaiah et al.*, 1975; *Walton*, 1980].

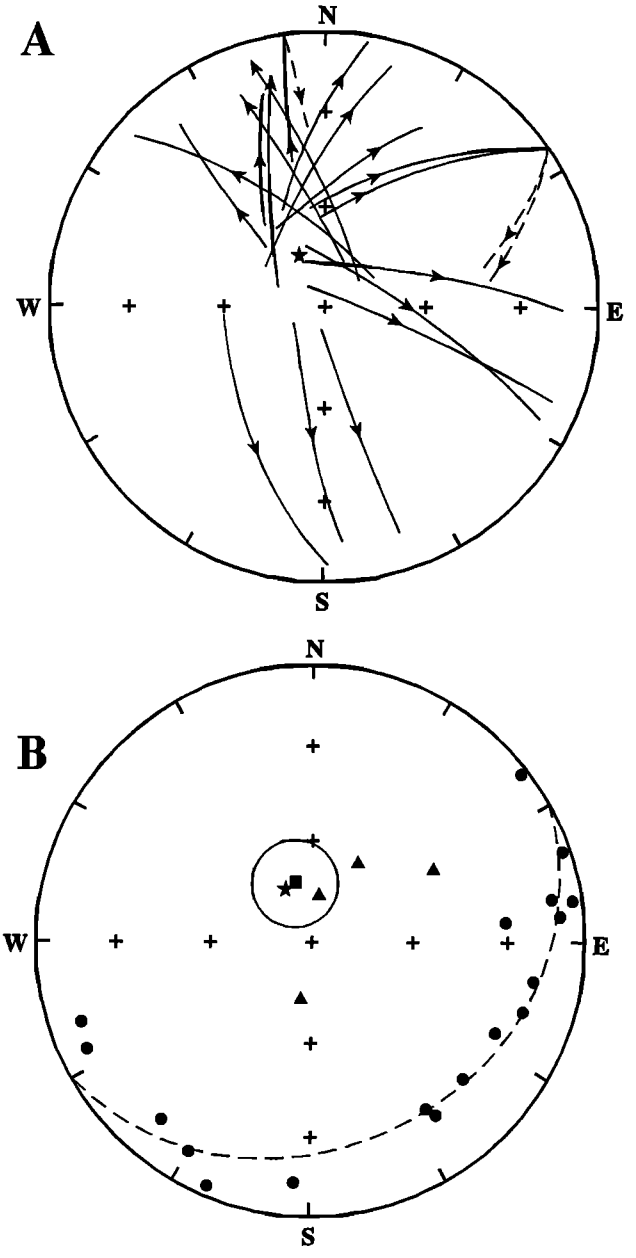


Fig. 9. Stereograms. (*a*) NRM vector trajectories upon AF and thermal demagnetization for specimens from the conglomerate clasts. Star is the present earth's magnetic field direction. (*b*) The normals to remagnetization planes shown in (*a*) with the best-fitting great circle through the normals (dashed line), and the mean direction (square) with its cone of 95% confidence obtained by combining the normals (dots) with the stable endpoints (triangles) using the method of *Bailey and Halls* [1984].

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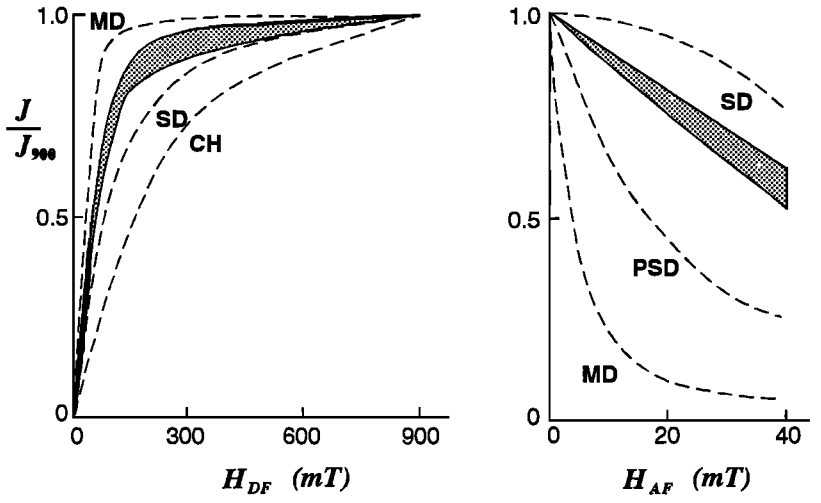


Fig. 10. Saturation isothermal remanence magnetization (SIRM) curves for 11 specimens from the ore and adjacent host rocks (*a*) SIRM acquisition with all curves in the shaded area, and (*b*) AF demagnetization of SIRM with all curves in the shaded area. The type curves are for single-domain (SD), pseudo-single domain (PSD), and multidomain (MD) magnetite, and for coarse-grained hematite (CH).

Therefore the CAI and fluid inclusion data rule out the possibility that an original magnetite magnetization in the sphalerite such as an Ordovician diagenetic remanence may have been reset as a thermal viscous remanent magnetization in the Middle-Late Devonian, and therefore they also indicate that the remagnetization event is coeval with the mineralization.

Trace MVT mineralization is widespread in Cambrian to Ordovician strata of the northern Humber Zone [*Saunders and Strong*, 1986; *Saunders et al.*, 1992]. These traces are found either as (1) fracture fillings in dissolution collapse breccias or pseudobreccias on a large scale, or, as (2) clots, disseminations, and fragment coatings on a limited scale. *Prasad* [1986] studied the paleomagnetism of the St. George limestone in an area about 40 km north of the NZM. He found almost the same ChRM direction as at the NZM and suggested from its pole that the limestone may have been remagnetized after the Early Ordovician. Our similar ChRM direction suggests that the pervasive trace MVT mineralization and remagnetization event in the northern Humber Zone area is cogenetic and coeval with that at the NZM.

An host rock remagnetization and MVT mineralization event near the Middle to Late Devonian boundary coincides temporally and spatially with the Acadian orogeny. It played an important role in forming the structural style of the Humber Zone where deformation is characterized by thrust-related structures which overprint the Taconic regional fabric of both the shelf carbonates and Grenville basement [*Cawood and Williams*, 1986, 1988]. Conversely, deformation during the subsequent Late Carboniferous-Permian Alleghenian orogeny is poorly developed, so that its effects are almost unrecognized [*Williams*, 1980; *Cawood and Williams*, 1988]. For example, the Early Carboniferous (Visean) rocks are undeformed in the Deer Lake basin of western Newfoundland where they overlap Acadian structures [*Hyde*, 1983]. Thus if any tectonic orogeny is involved in the emplacement of the MVT mineralization in the northern Humber Zone, it would likely be the Acadian rather than the Alleghenian orogeny.

Lane [1990] has extensively analyzed sedimentologic, stratigraphic, structural, petrographic and geochemical evidence in the NZM area. He found that the sphalerite and saddle dolomite overprint a complex series of dolomites and diagenetic limestones that were formed over a prolonged period from the Middle Ordovician to the Silurian. From an analysis of mineralized fractures and faults in the mine area, *Lane* [1990] suggested that the sphalerite ore is related to regional fracturing during the Acadian orogeny. Also, *Hall et al.* [1989] reported that ^{40}Ar-^{39}Ar age spectrum data for alkali feldspars from MVT mineralization in western Newfoundland indicate two thermochemical events, one at 370-350 Ma and the other at 210 Ma. They associated the older Late Devonian event with the sphalerite mineralization event which agrees with the age indicated by the NZM pole position.

Based on the geochemical similarity, *Kesler and van der Pluijm* [1990] hypothesized that several Appalachian MVT deposits including the NZM were formed coevally within the Middle Ordovician to Middle Devonian interval. They speculated that these deposits were formed when the Table Head group was remagnetized, which *Hodych* [1989] had associated with the Taconic orogeny. This study shows that the NZM magnetization is very different from that of the Table Head group and that, for the NZM at least, the mineralization and host rock remagnetization event is related to the later Acadian orogeny.

Models for MVT Ore Genesis

MVT deposits in the midcontinent are thought to be related to the migration of hot brine fluids out of deep foreland basins into the shallow marine carbonates of the foreland when the basins were deformed and uplifted during the Alleghenian or Ouachita orogenies in the late Paleozoic [*Garven and Freeze*, 1984*a*, *b*; *Leach and Rowan*, 1986; *Oliver*, 1986; *Duane and de Wit*, 1988; *Clendenin and Duane*, 1990; *Bethke and Marshak*, 1990]. Paleomagnetic studies on MVT deposits in southeast Missouri [*Wisniowiecki et al.*, 1983], east Tennessee [*Bachtadse et al.*, 1987], northern Arkansas and Tri-State [*Pan et al.*, 1990], and central Missouri [*Symons and Sangster*, 1991] show that these MVT deposits and host rocks acquired their ChRM during Kiaman reversed superchron. Their geologic and magnetic characteristics, combined with magnetic tests on MVT ore stage minerals, indicate that the host rock remagnetization is essentially coeval with the mineralization magnetization [*Symons and Sangster*, 1993], linking their genesis to either the Alleghenian or Ouachita orogeny. Similarly the MVT Zn-Pb deposit and host rocks at Polaris in the Canadian Arctic Archipelago retain a Late Devonian pole. A paleomagnetic fold test shows that their magnetization is prefolding and associated with the adjacent Late Devonian Ellesmerian orogeny [*Symons and Sangster*, 1992]. Applying this model, it appears that Acadian orogenic deformation and uplift forced hot basinal brine fluids into the porous and fractured shelf carbonates of the northern Humber Zone, resulting in remagnetization of the host rocks and deposition of MVT mineralization.

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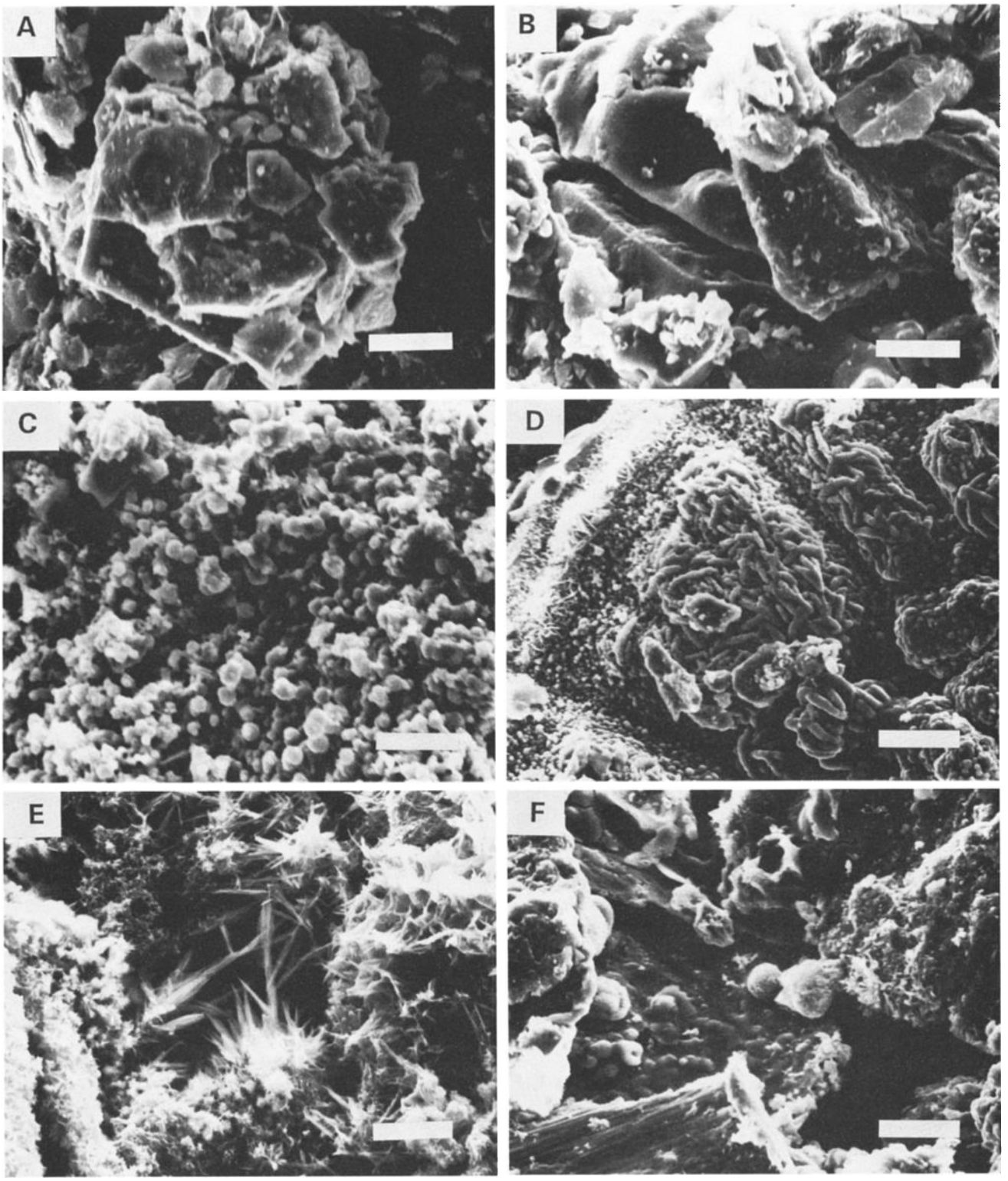


Fig. 11. Scanning electron micrographs of magnetic extracts from (*a*) ore, scale bar is 10 μm; (*b*) dolostone, 9 μm; (*c*) syndepositional breccia clast, 2 μm; (*d*) ore, 4 μm; (*e*) ore, 4 μm; (*f*) ore, 3 μm.

CONCLUSIONS

1. The sphalerite ore and Early Ordovician host rocks of the Newfoundland Zinc Mines MVT deposit in the Humber Zone carry similar normal-polarity ChRM directions upon AF or thermal demagnetization. When the site mean directions are averaged, they give a mean ChRM direction of 9°/-30° (declination/inclination) (N = 16 sites, α_{95} = 6°, k = 46).

2. SIRM and thermal step demagnetization results show that the ChRM in both the ore and host rocks is carried mostly by SD with some PSD magnetite. SEM-EDA observations reveal that the magnetite has

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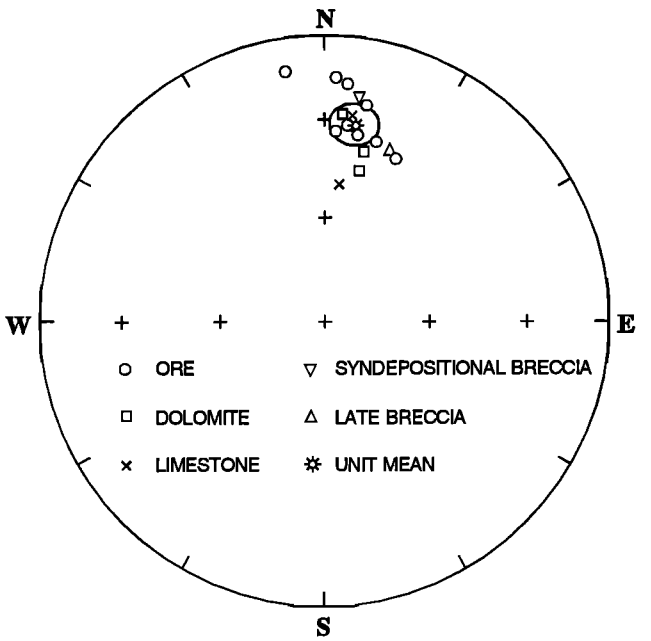


Fig. 12. Stereogram of the site mean directions for various lithologies. The unit mean direction is circumscribed by its cone of 95% confidence.

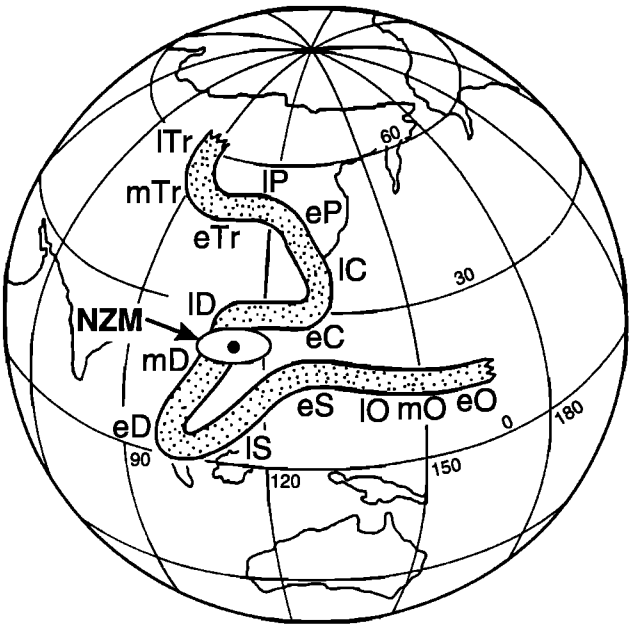


Fig. 14. The NZM pole position with its oval of 95% confidence relative to the apparent polar wander path of *Van der Voo* [1990].

authigenic characteristics, indicating that it is secondary and probably formed in a chemical process.

3. Two paleomagnetic breccia tests are negative. They require that the ChRM in the host rocks be a secondary remagnetization that postdates the early Middle Ordovician, and that this remagnetization event occurred during or after the MVT mineralization event. This is supported by the occurrence of trace MVT mineralization in allochthonous Middle Ordovician rocks that accreted during the Early to Middle Ordovician Taconic orogeny.

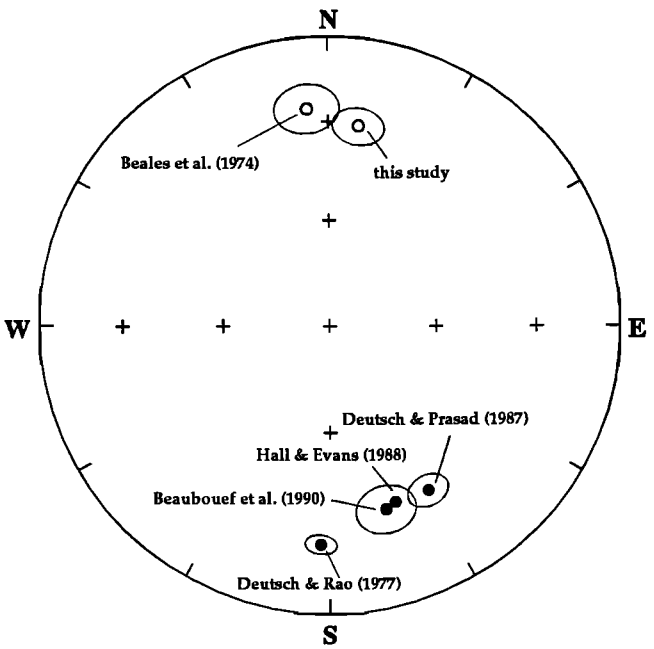


Fig. 13. Stereogram of mean directions reported for the St. George group circumscribed by their cones of 95% confidence (α_{95}). *Hall and Evans* [1988] did not report an α_{95} value.

4. The mean ChRM component gives a pole position at 22°N, 113°E ($dp = 3°, dm = 6°$) that straddles the Middle to Late Devonian boundary on the North American APW path, indicating an age of about 374 ± 10 Ma. The determined paleoinclination shows that the acquired magnetization and, from conclusion 3 above, the mineralization cannot be younger than Late Devonian.

5. Resetting of an early ChRM by a thermochemical event within the available Middle Ordovician to Late Devonian time span is precluded by the combination of low paleotemperatures of <200°C indicated by the CAI and fluid inclusion data for ore emplacement [*Nowlan and Barnes*, 1987; *Lane*, 1990], and by unblocking temperatures of >550°C for the magnetite carrying the ChRM in some ore and host rock specimens.

6. We interpret our data to show that the MVT ore at NZM acquired its ChRM during ore genesis, and that its host rocks were remagnetized in the same thermochemical event at about 374 ± 10 Ma ago. This event coincides in time with the Acadian orogeny in the study area. Support for this age comes from (1) ^{40}Ar-^{39}Ar age spectrum data [*Hall et al.*, 1989]; (2) trace MVT mineralization in Acadian fractures but not in the younger late Paleozoic Alleghenian fractures; and (3) regional paleomagnetic studies of Early to Middle Ordovician carbonate rocks in the Humber Zone, which isolate a comparable normal-polarity ChRM direction only where trace MVT mineralization is also found [*Prasad*, 1986; *Saunders and Strong*, 1986; *Saunders et al.*, 1992].

7. Finally, in tying ChRM acquisition by the mineralization at NZM and the remagnetization ChRM of its host rocks to the Acadian orogeny, this study supports the results of several recent paleomagnetic studies on MVT deposits elsewhere in the Appalachians, midcontinent of the United States, and Arctic Archipelago of Canada that relate the genesis of the MVT mineralization to an adjacent orogenic event.

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help in collecting, description of the specimens, and comments on the manuscript. We are also grateful to J. W. Robinson of the University of Windsor who ran the SEM. The manuscript benefitted from the many useful comments of J. T. Hagstrum, J. W. Geissman, and an anonymous journal reviewer. This study was funded by an NSERC operating grant to D.T.A.S.

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